

COMPASS_Synoptic_LE_Soil-Fluxes_2023_CH4

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Table of contents

0.1	Goals of this code:	2
0.2	Set Up & Load Packages	2
0.3	Pull in Smart Chamber Files from the Raw Data folder	2
0.4	Unfortunately - we had many problems with labels - we need to include the sites in the Plot_ID	2
0.5	Create a unique Plot_ID, check if there is a meta-data file, if not, create a meta-data file	2
0.6	Mark fluxes that were retaken - this is the first issue! Need to get all but last flux to be TRUE for drop - fieldnotes with explanation for flux retaken - usually weird zeros, Steven probe not inserted in the soil, or label mistake	2
0.7	Match up the metadata to the rest of the data	3
0.8	fixing some Deadbands Revisit here, it is not working to fix the dead band . . .	3
0.9	Change the units from ppb to nmol	3
0.10	Calculate fluxes	4
0.11	Visualize fluxes and sort by r2	4
0.12	check the fluxes for which I changed the start time due to lags or ebullition (recorded in notes of metadata)	4
0.13	Spot check a few fluxes with R2 > 0.9	16
0.14	Take a look at the fluxes <0.9 and a flux estimate below 1	19
0.15	Take a look at the bad fluxes <0.9 and have a high flux estimate	22
1	See if we want to remove these fluxes or if we want to try and recalculate these ones?	22
1.1	Look at the really high or really low ones to see if there is a timing issue or something	24
1.2	Check all fluxes	27

1.3	QAQC - Flags	30
1.4	Remove any fluxes that do not fit criteria	30
1.5	Do some visualization of QAQC plots to assess fluxes	32
1.6	Read out final dataframe and export for use elsewhere	34

0.1 Goals of this code:

1. Reads in raw smart chamber
2. If no metadatafile, write one out & fields will be: Label the IDs pulled from the label, Plot *fixed so that if there are any issues in the label they are good here Drop column – false as long as label looks right adjust time – would be initially populated by the start time in the OR blank adjust observation length – initially populated by the obs length OR blank notes / comment
3. Code reads in the meta data and applies it to the data
 - a. i.e. drops fluxes that have the drop column TRUE or FLAG this
 - b. adjusting times as needed
4. Now you compute fluxes
5. Visualize and look for issues
6. If found, report to meta data file then go back to 3 and reanalyze
7. Apply algorithmic qaqc and generate flags

0.2 Set Up & Load Packages

0.3 Pull in Smart Chamber Files from the Raw Data folder

0.4 Unfortunately - we had many problems with labels - we need to include the sites in the Plot_ID

0.5 Create a unique Plot_ID, check if there is a meta-data file, if not, create a meta-data file

```
[1] "CSV file exists and has been read into the code."
```

0.6 Mark fluxes that were retaken - this is the first issue! Need to get all but last flux to be TRUE for drop - fieldnotes with explanation for flux retaken - usually weird zeros, Steven probe not inserted in the soil, or label mistake

##Dont need to pull L1 temp data or any other temps like we do for macrophyte because the smartchamber records the temp

0.7 Match up the metadata to the rest of the data

```
YYYY-MM-DD format failed for start_dates; trying MM/DD/YYYY
```

```
1 entry had no timestamp matches!
```

```
YYYY-MM-DD format failed for start_dates; trying MM/DD/YYYY
```

```
YYYY-MM-DD format failed for start_dates; trying MM/DD/YYYY
```

0.8 fixing some Deadbands Revisit here, it is not working to fix the dead band

```
Warning in left_join(., time_overrides, by = "Plot_ID"): Detected an unexpected many-  
to-many relationship between `x` and `y`.
```

```
i Row 30585 of `x` matches multiple rows in `y`.
```

```
i Row 16 of `y` matches multiple rows in `x`.
```

```
i If a many-to-many relationship is expected, set `relationship =  
  "many-to-many"` to silence this warning.
```

```
##Extracting initial CH4 concentration
```

0.9 Change the units from ppb to nmol

```
#change the units  
alldat_trimmed$CH4_nmol <- ffi_ppb_to_nmol(alldat_trimmed$ch4,  
                                           volume = (alldat_trimmed$TotalVolume_fixed_cm3 / 1000000), #  
                                           temp = alldat_trimmed$chamber_t)      # degrees C
```

```
Assuming atm = 101325 Pa
```

```
Using R = 8.31446261815324 m3 Pa K-1 mol-1
```

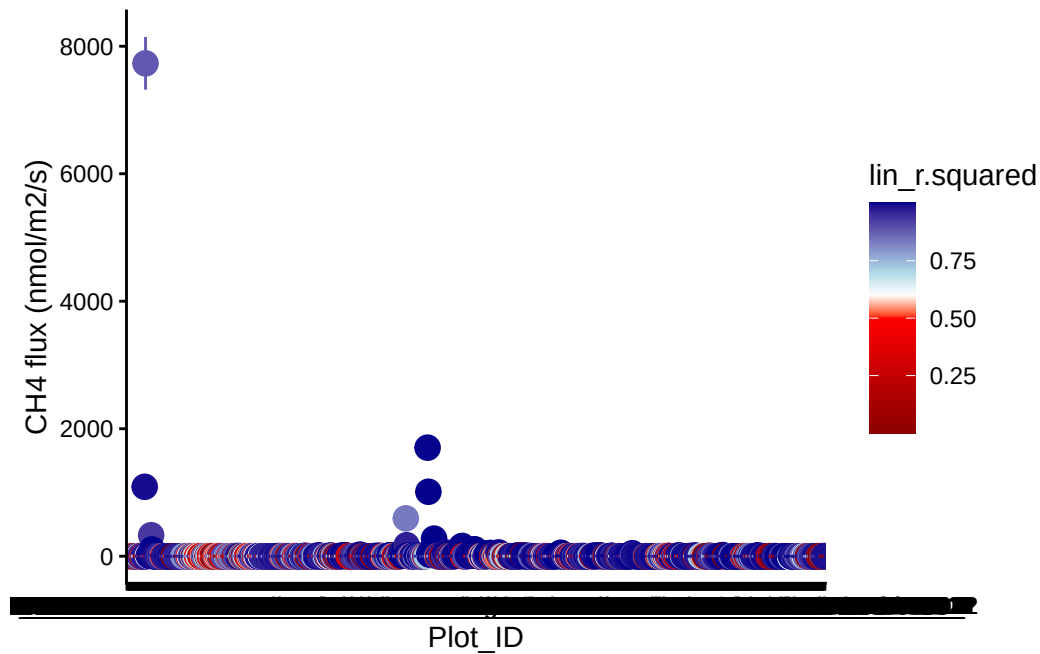
```
#> Assuming atm = 101325 Pa
```

```
#> Using R = 8.31446261815324 m3 Pa K-1 mol-1
```

```
alldat_trimmed$CH4_nmol_m2 <- alldat_trimmed$CH4_nmol / 0.0318 #adjust for the area
```

0.10 Calculate fluxes

0.11 Visualize fluxes and sort by r2



0.12 check the fluxes for which I changed the start time due to lags or ebullition (recorded in notes of metadata)

```
# grab trim limits for each overridden Plot
trim_limits <- alldat_trimmed %>%
  filter(Plot_ID %in% time_overrides$Plot_ID) %>%
  group_by(Plot_ID) %>%
  summarise(
    t_start_trim = min(TIMESTAMP),
    t_end_trim   = max(TIMESTAMP),
    .groups = "drop"
  )

# use full alldat + trim limits to classify each point
r_all <- alldat %>%
  filter(Plot_ID %in% time_overrides$Plot_ID) %>%
  left_join(trim_limits, by = "Plot_ID") %>%
```

```

group_by(Plot_ID) %>%
mutate(
  time_sec = as.numeric(difftime(TIMESTAMP, min(TIMESTAMP), units = "secs")),
  point_status = case_when(
    TIMESTAMP < t_start_trim ~ "Discarded (DeadBand)",
    TIMESTAMP > t_end_trim   ~ "Discarded (stop trim)",
    TRUE                     ~ "Used"
  )
) %>%
ungroup()

# ---- PLOTTING: one plot per Plot_ID (PDF friendly) ----

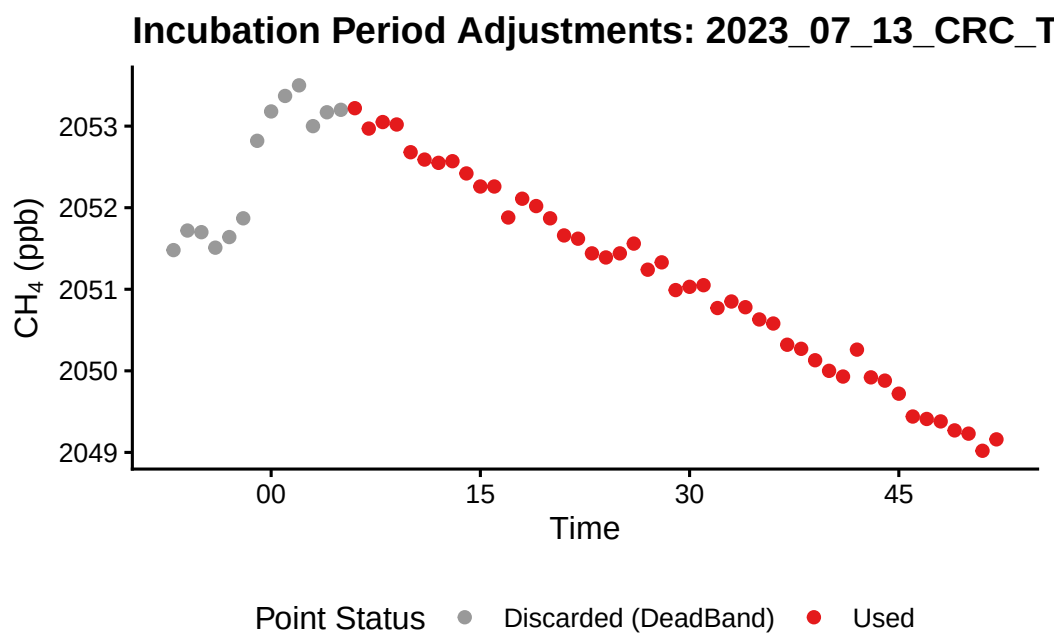
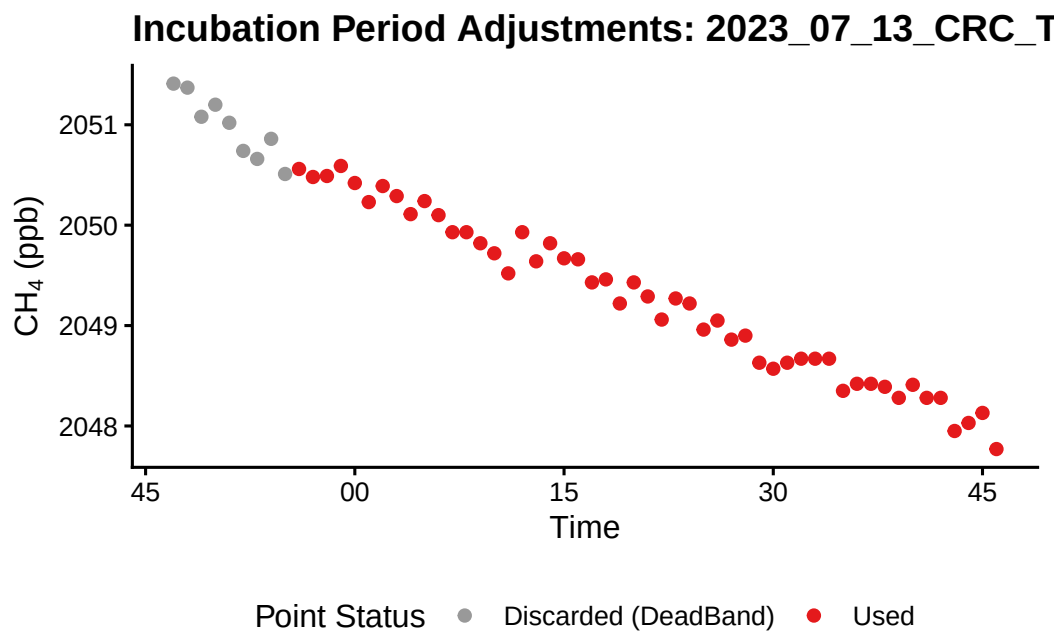
plot_ids <- unique(r_all$Plot_ID)

for (P in plot_ids) {

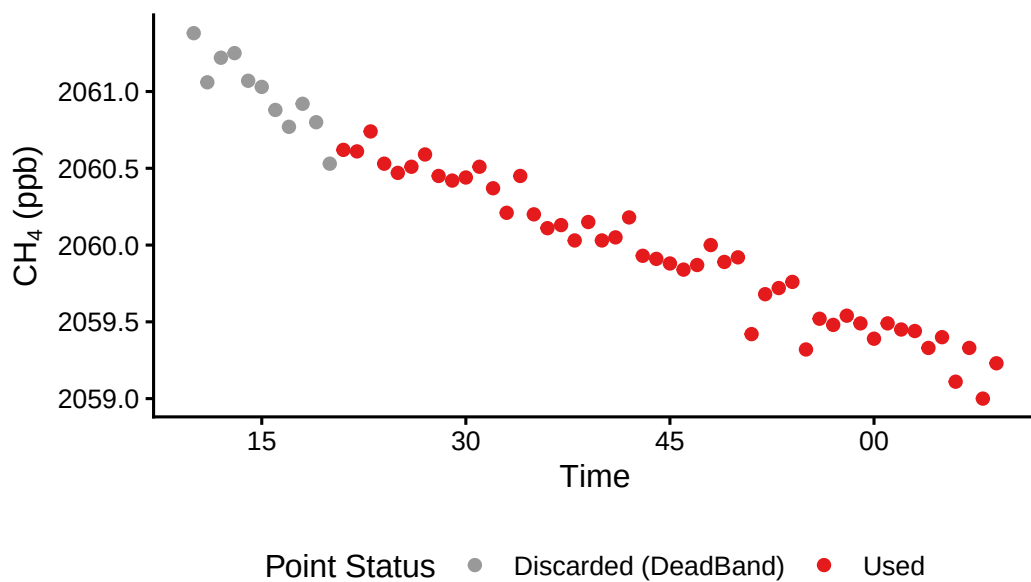
  p <- ggplot(
    r_all %>% filter(Plot_ID == P),
    aes(x = TIMESTAMP, y = ch4, color = point_status)
  ) +
    geom_point(size = 1.8) +
    scale_color_manual(
      values = c(
        "Used" = "#E41A1C",
        "Discarded (DeadBand)" = "grey60",
        "Discarded (stop trim)" = "grey20"
      )
    ) +
    theme_classic(base_size = 12) +
    ylab(expression(paste(CH[4], " (ppb)"))) +
    xlab("Time") +
    labs(
      title = paste("Incubation Period Adjustments:", P),
      color = "Point Status"
    ) +
    theme(
      legend.position = "bottom",
      plot.title = element_text(face = "bold")
    )

  print(p)  # IMPORTANT: each print() = new page in PDF
}

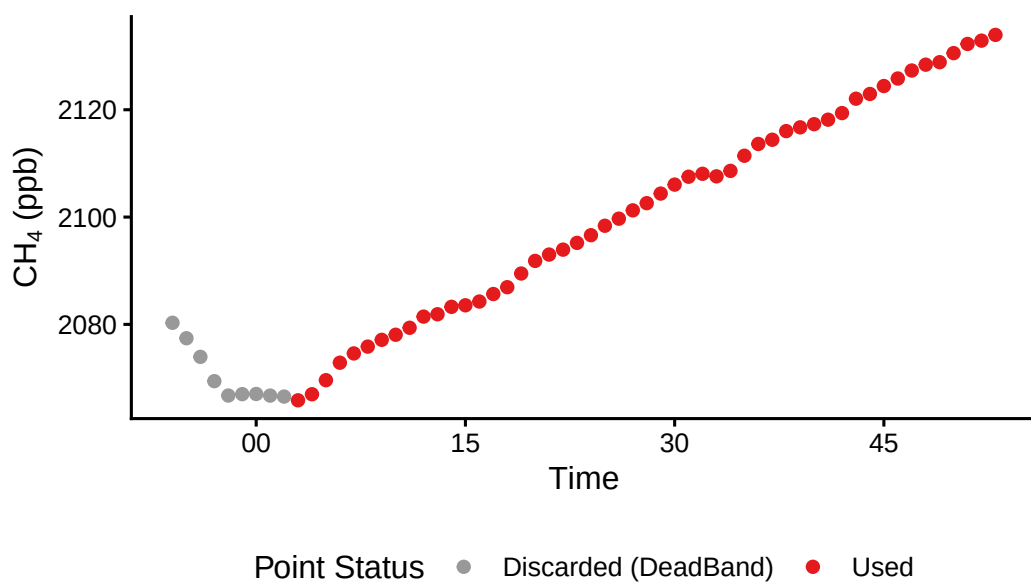
```



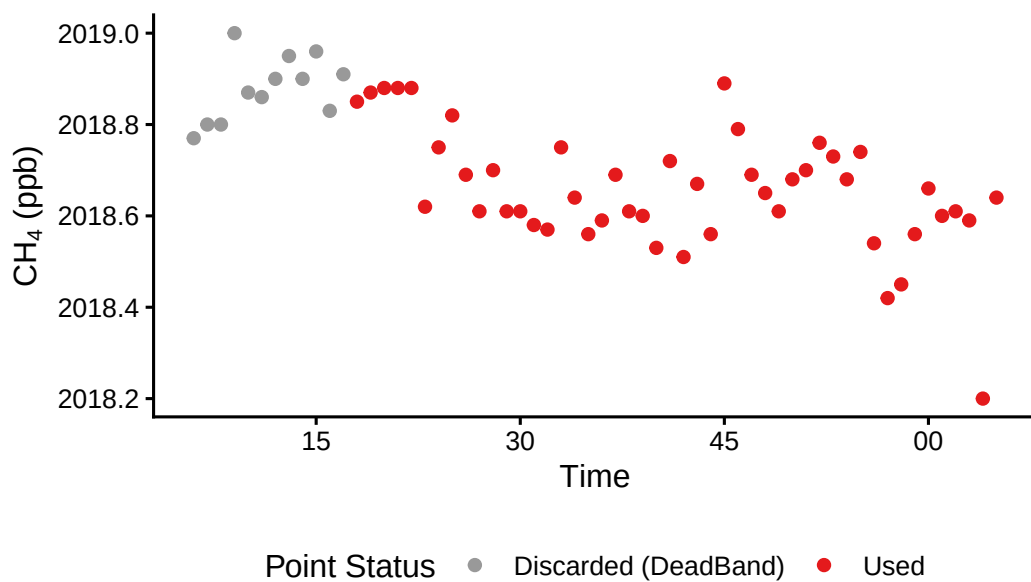
Incubation Period Adjustments: 2023_07_13_CRC_



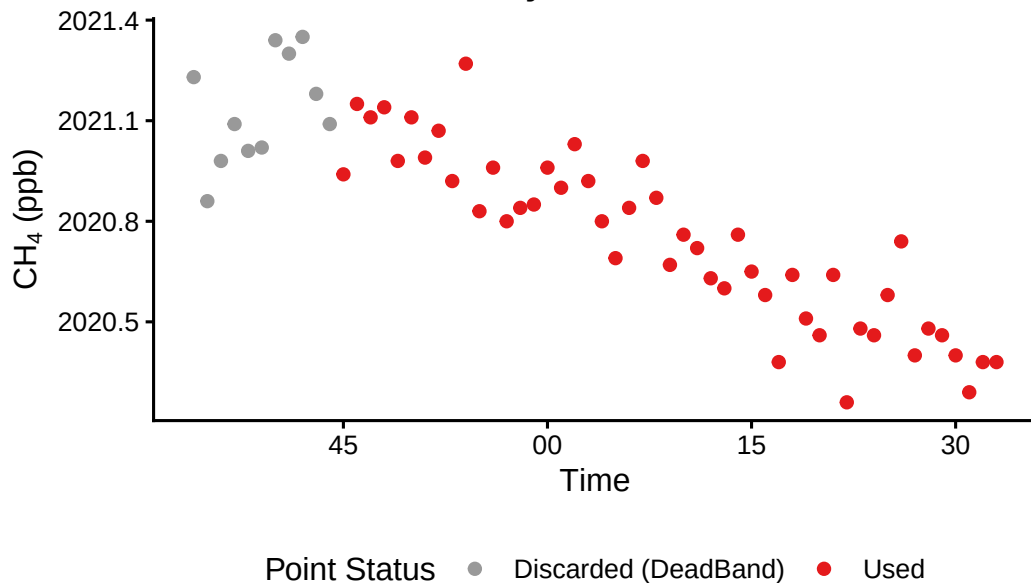
Incubation Period Adjustments: 2023_07_13_CRC_T

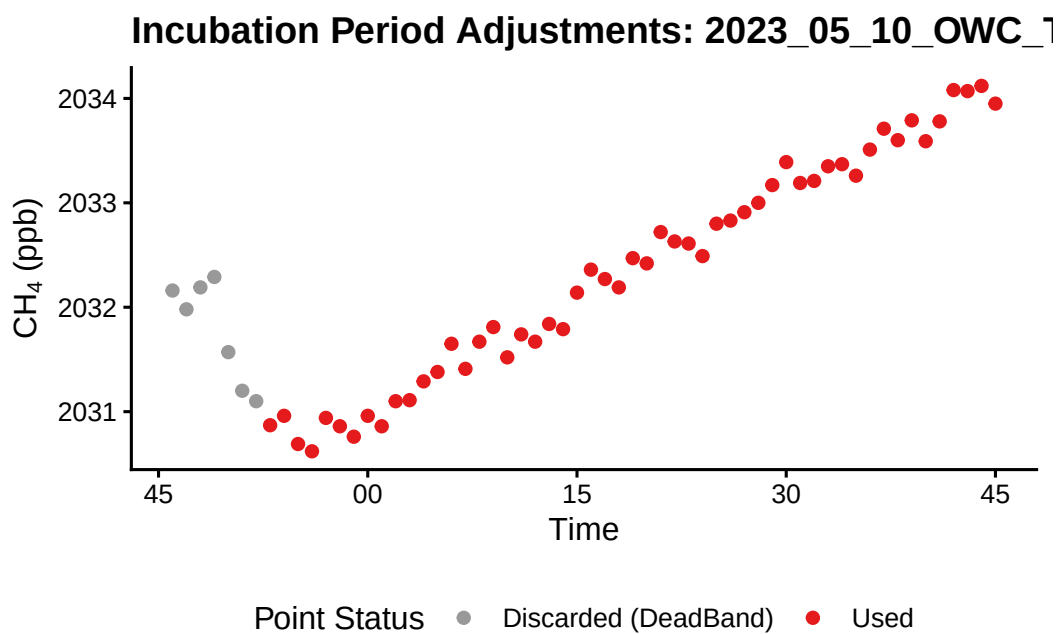
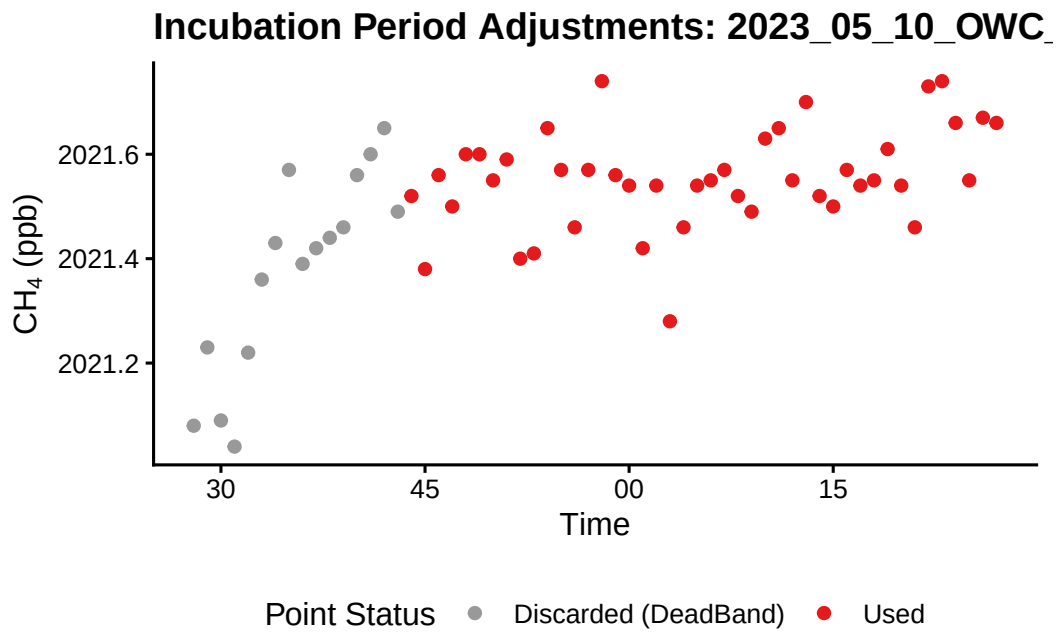


Incubation Period Adjustments: 2023_01_05_OWC_

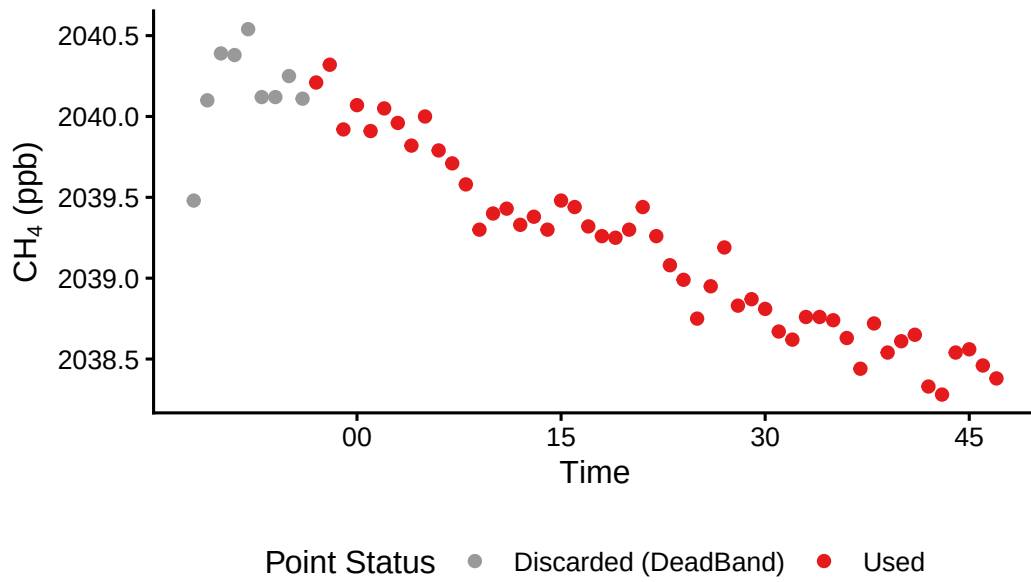


Incubation Period Adjustments: 2023_01_05_OWC_

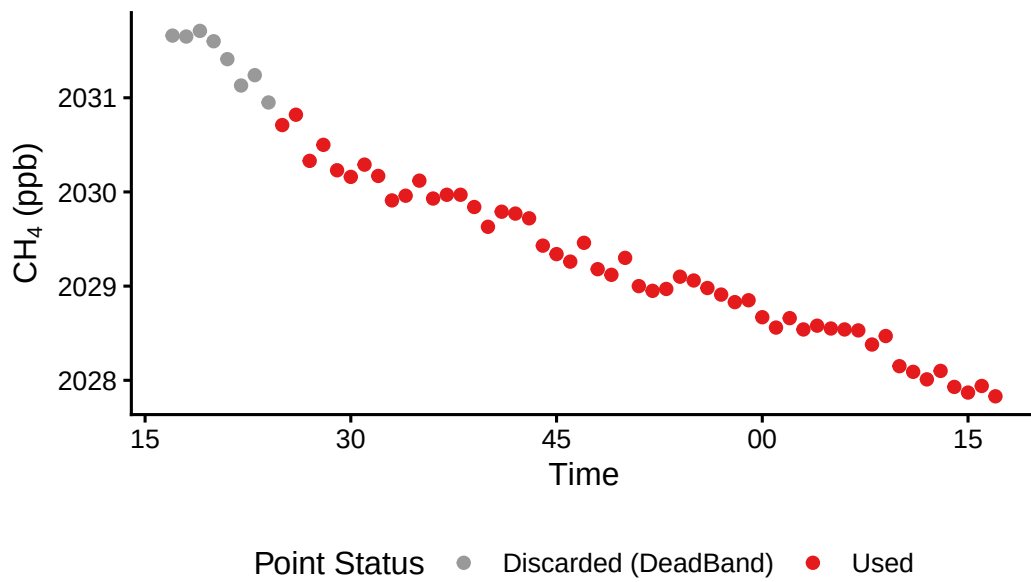


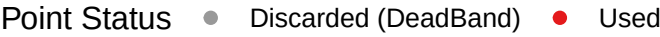


Incubation Period Adjustments: 2023_06_22_OWC_

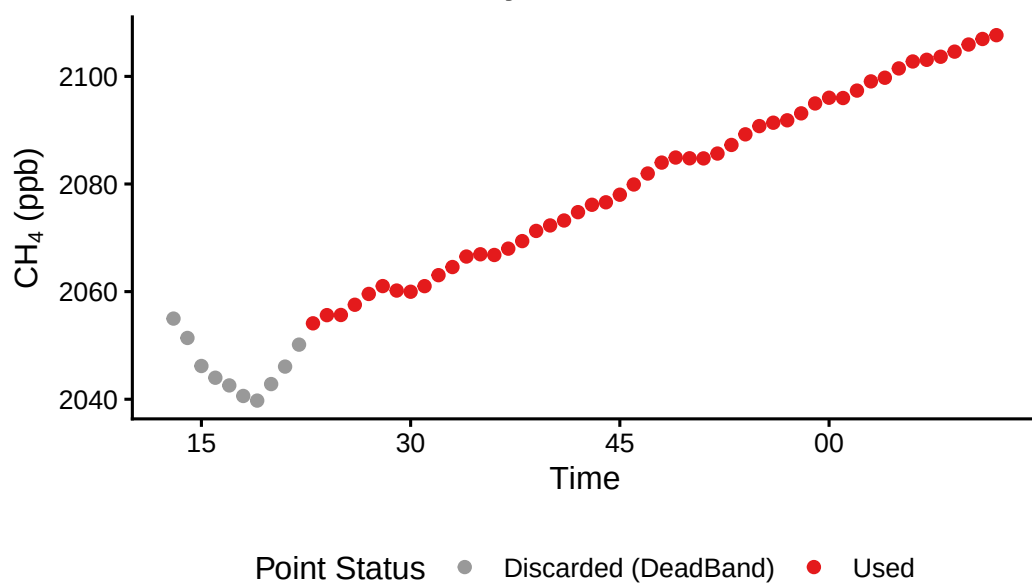


Incubation Period Adjustments: 2023_06_22_OWC_l

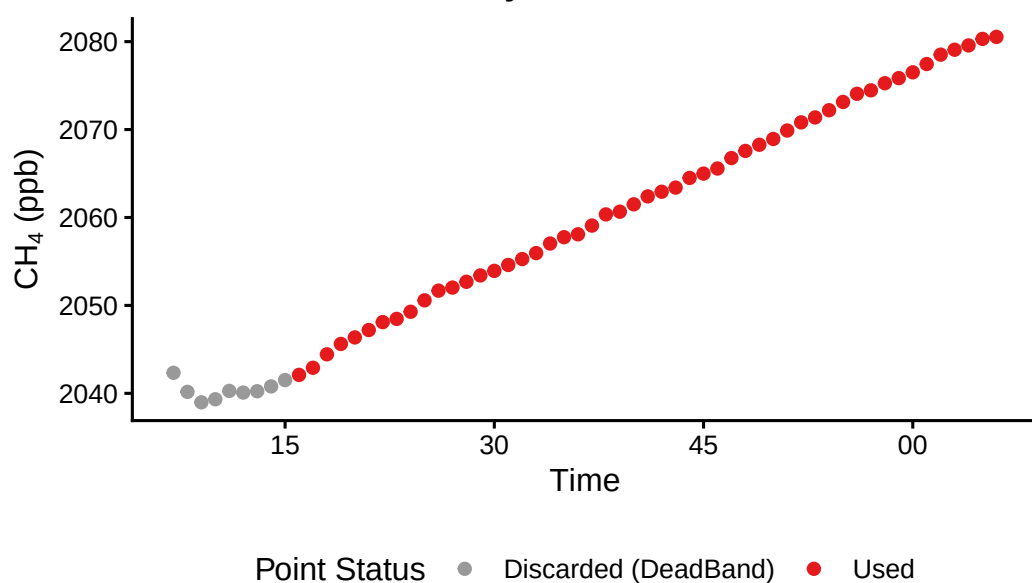




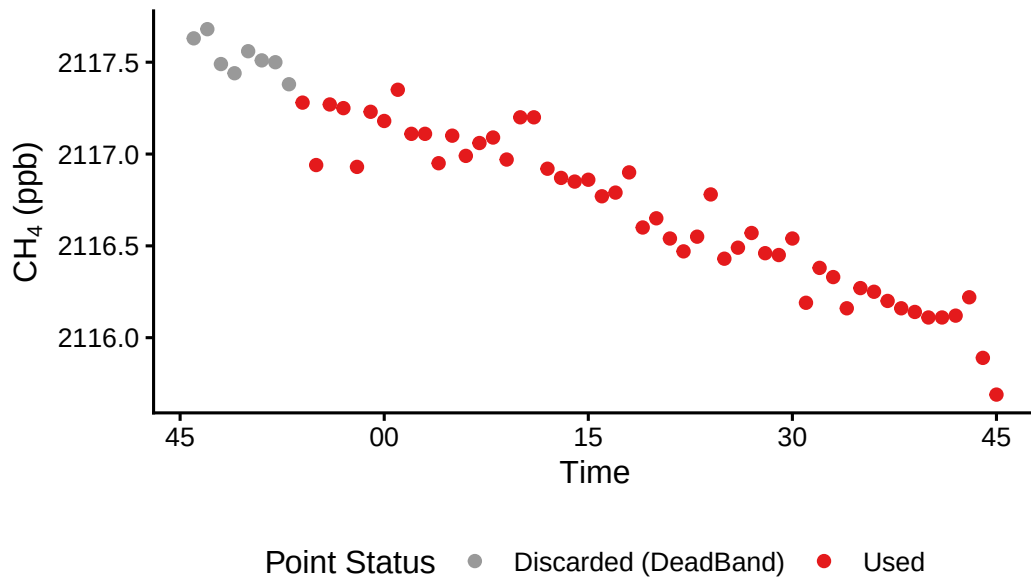
Incubation Period Adjustments: 2023_06_22_OWC_1



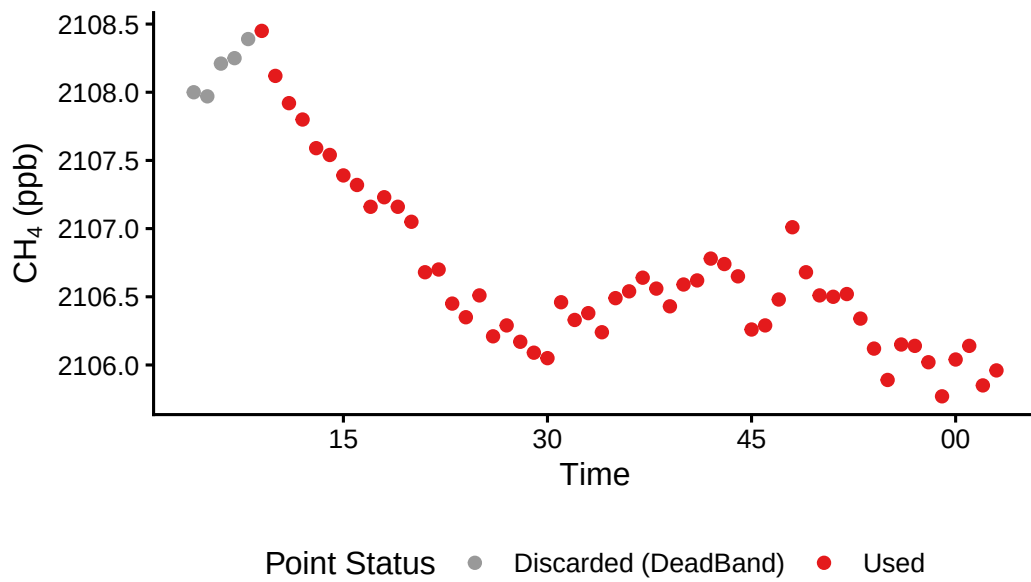
Incubation Period Adjustments: 2023_06_22_OWC_1

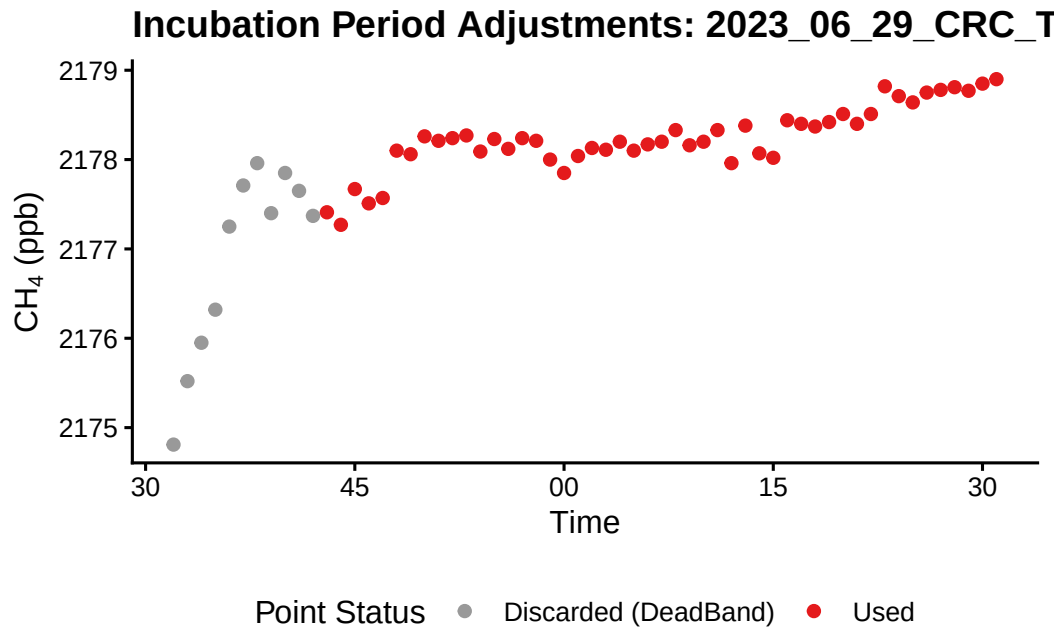


Incubation Period Adjustments: 2023_06_29_CRC_



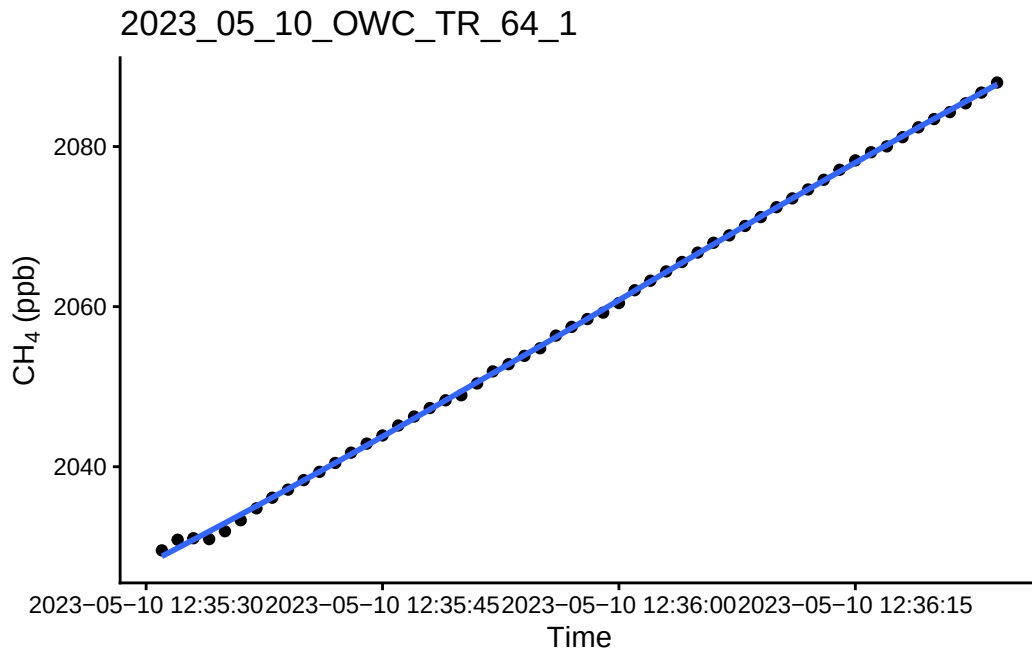
Incubation Period Adjustments: 2023_06_29_CRC_



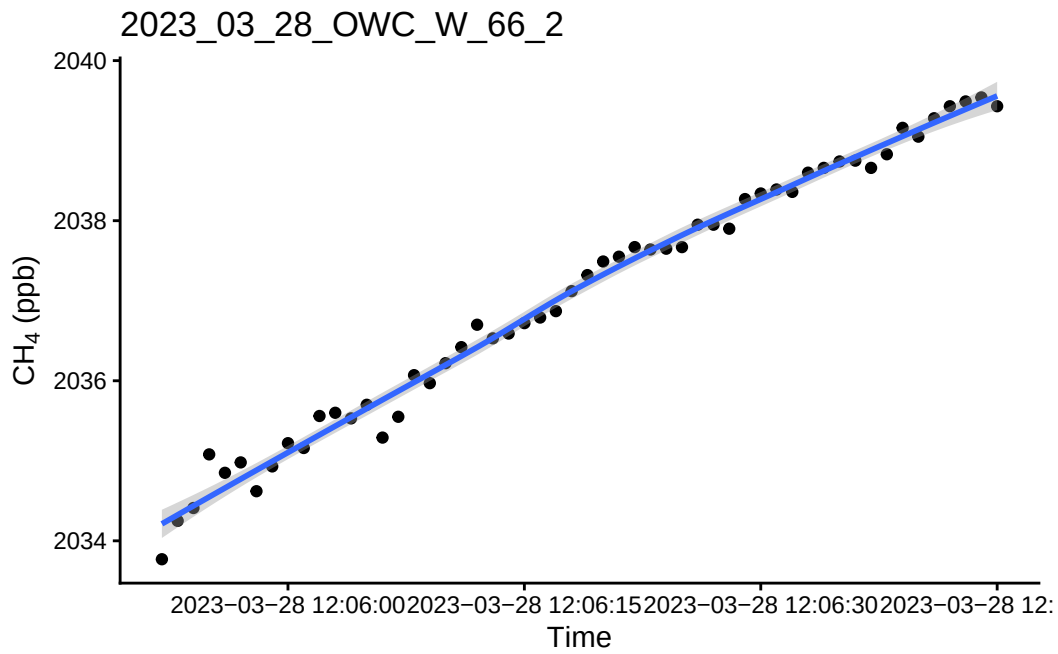


0.13 Spot check a few fluxes with $R^2 > 0.9$

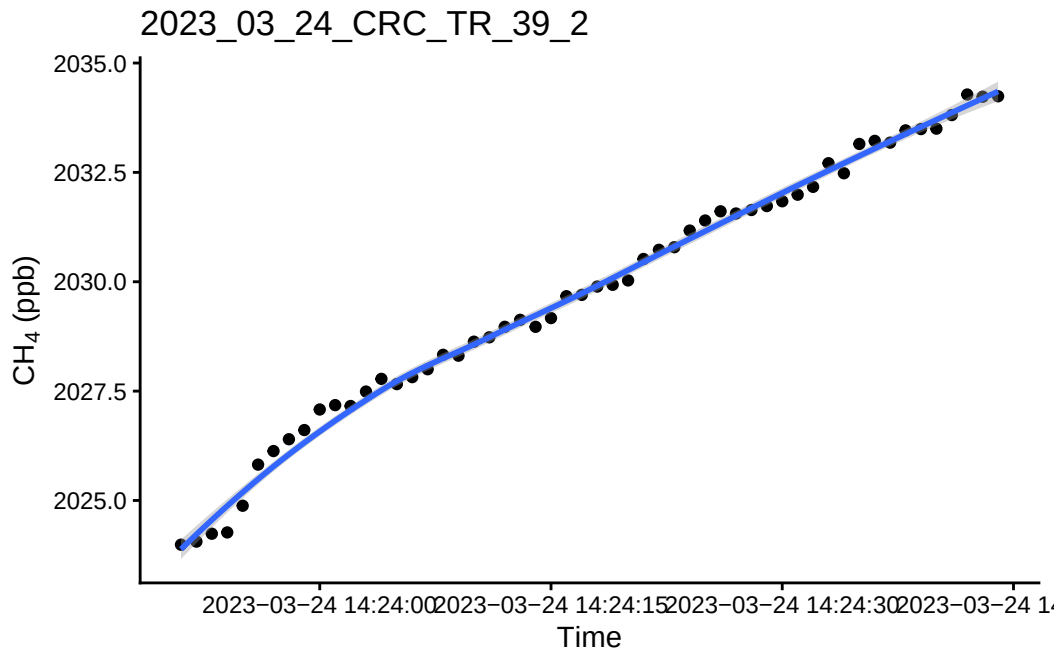
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



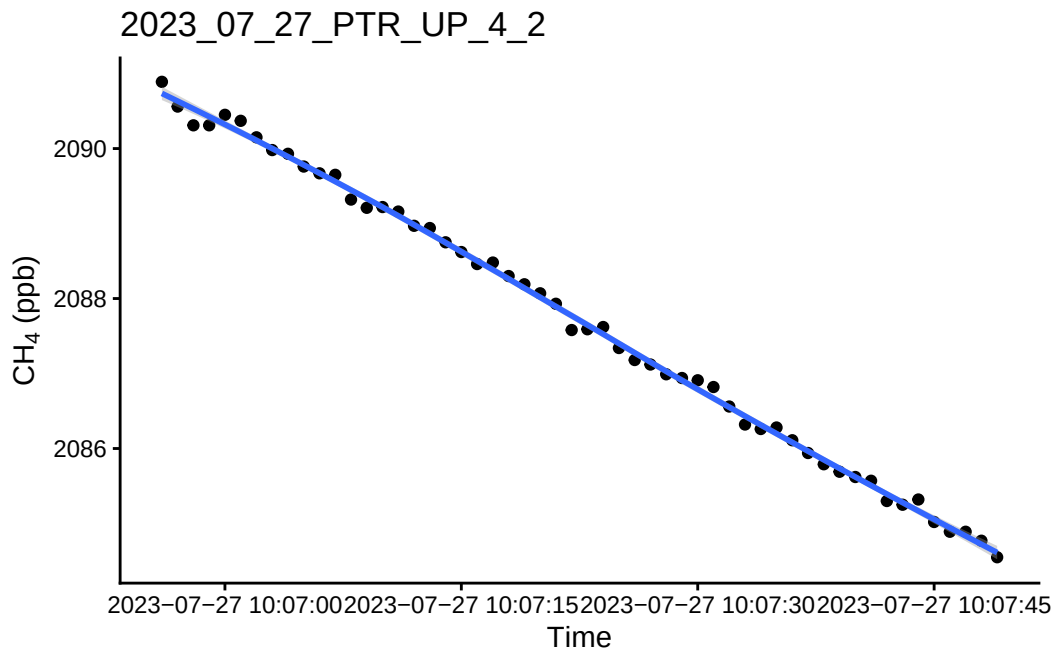
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



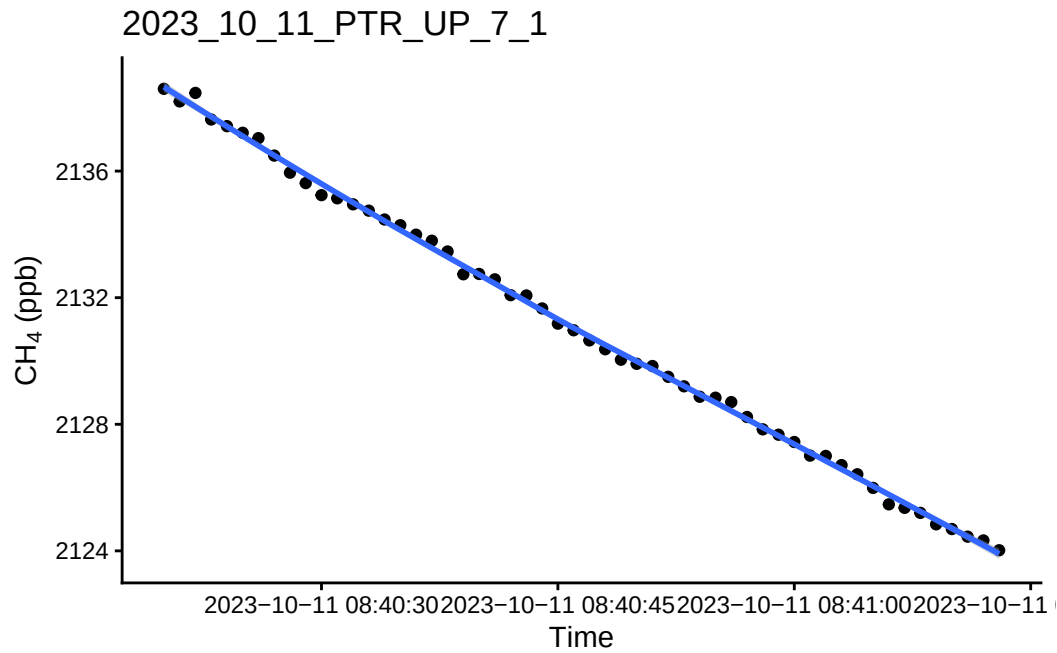
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

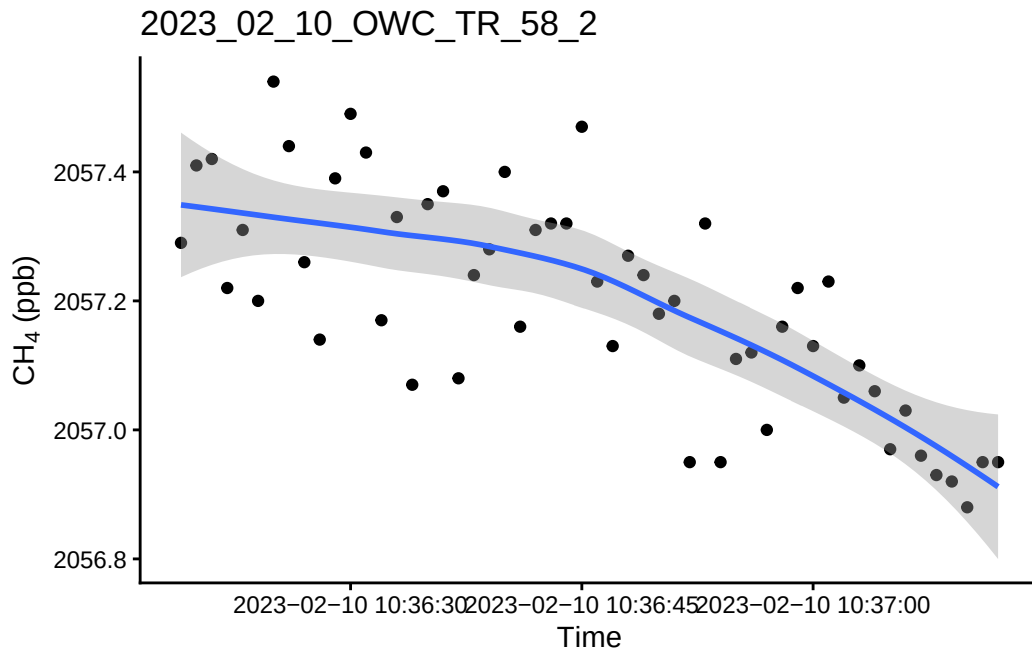


``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

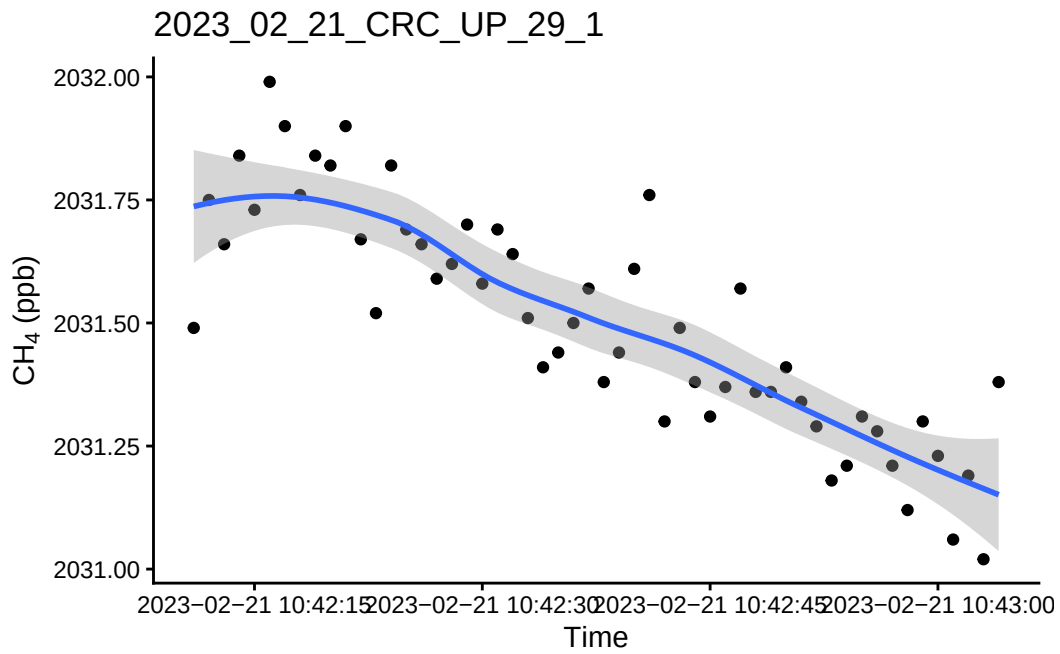


0.14 Take a look at the fluxes <0.9 and a flux estimate below 1

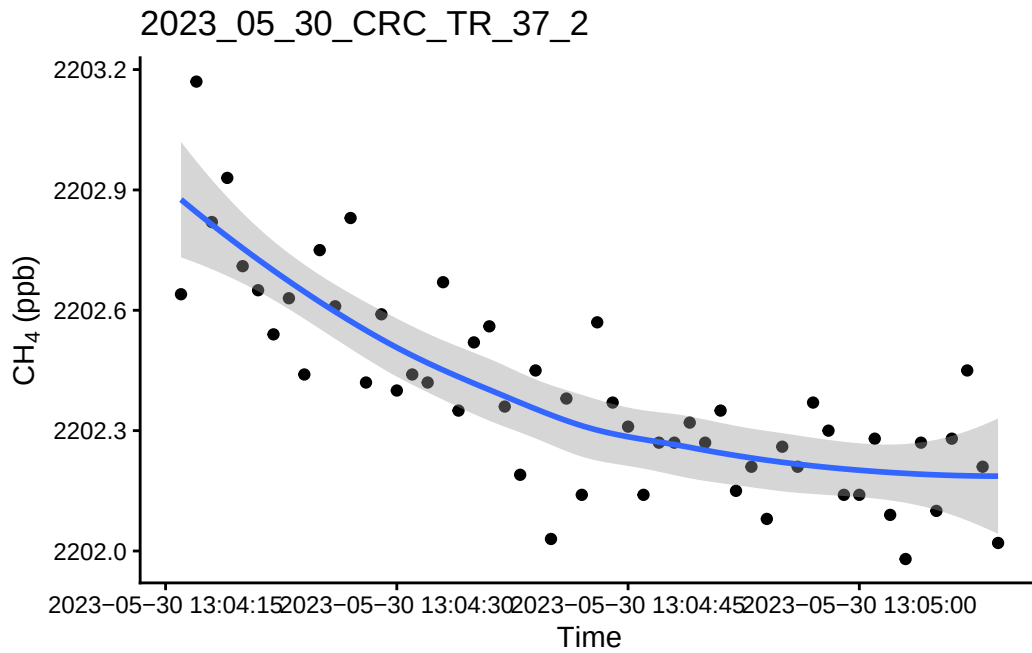
``geom_smooth()`` using `method = 'loess'` and `formula = 'y ~ x'`



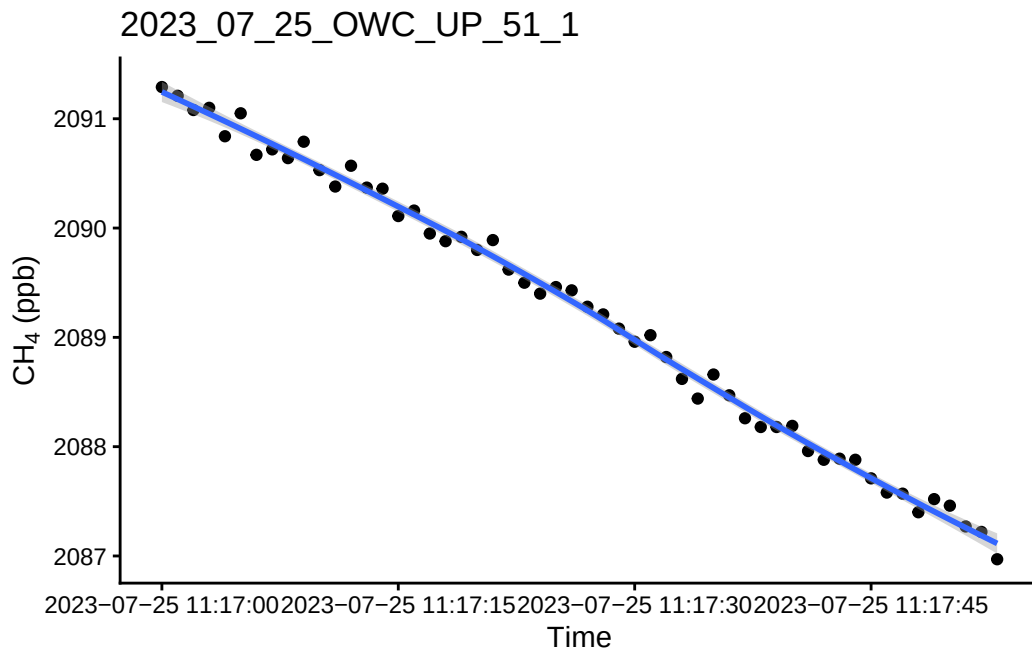
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



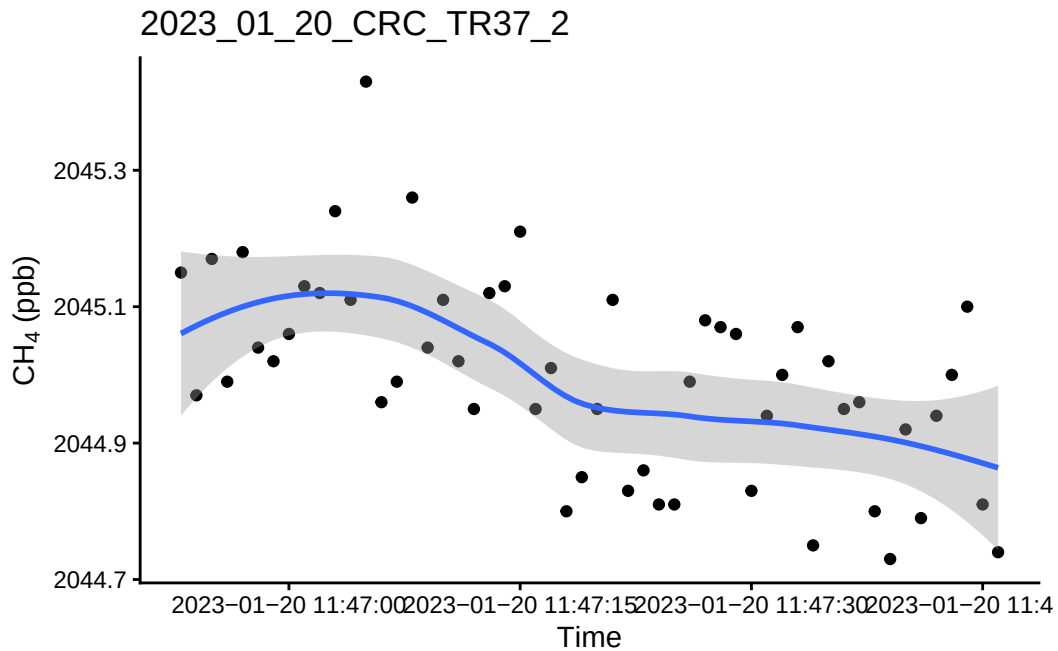
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



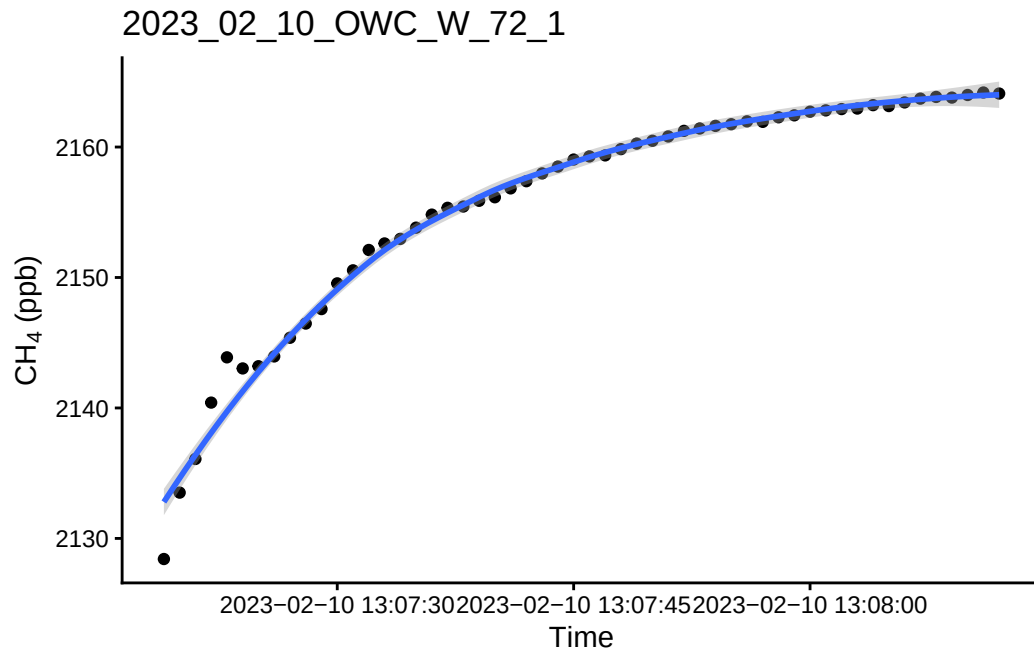
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



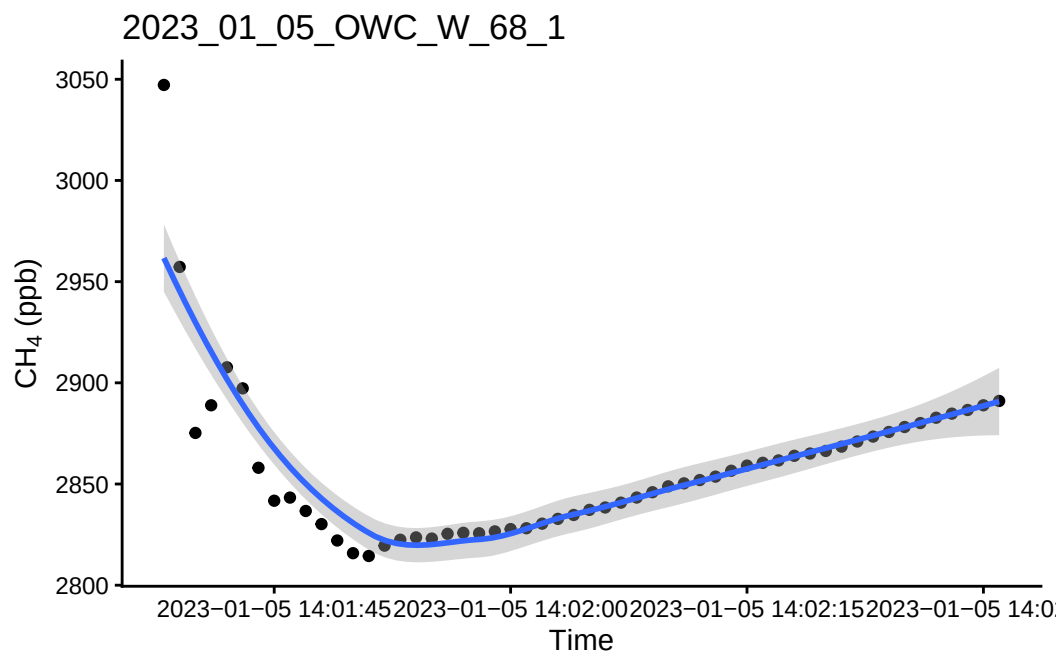
0.15 Take a look at the bad fluxes <0.9 and have a high flux estimate

1 See if we want to remove these fluxes or if we want to try and recalculate these ones?

``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

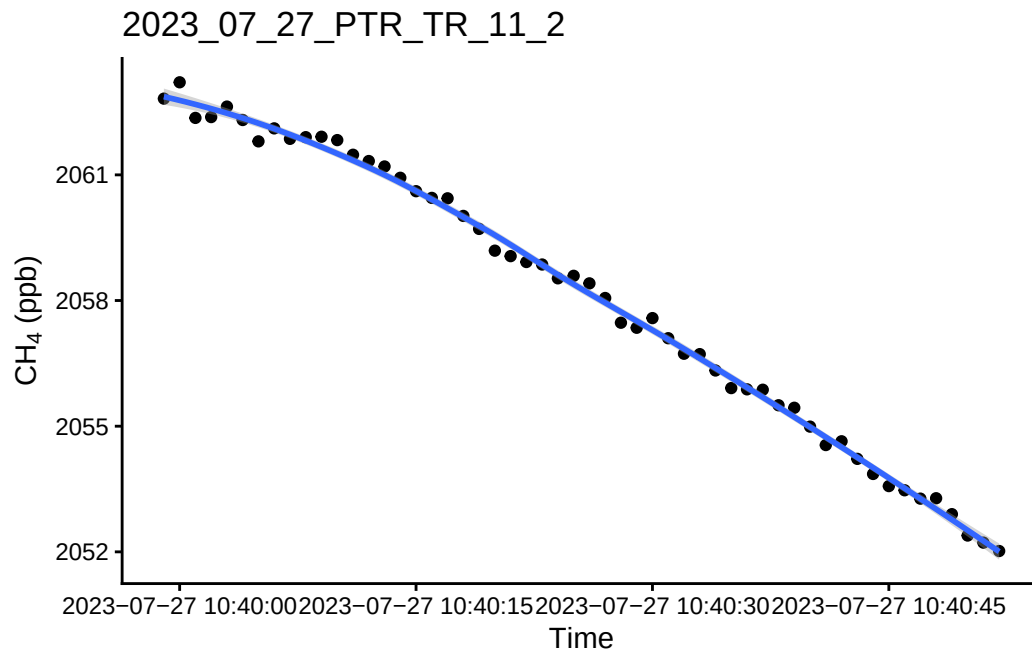


``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

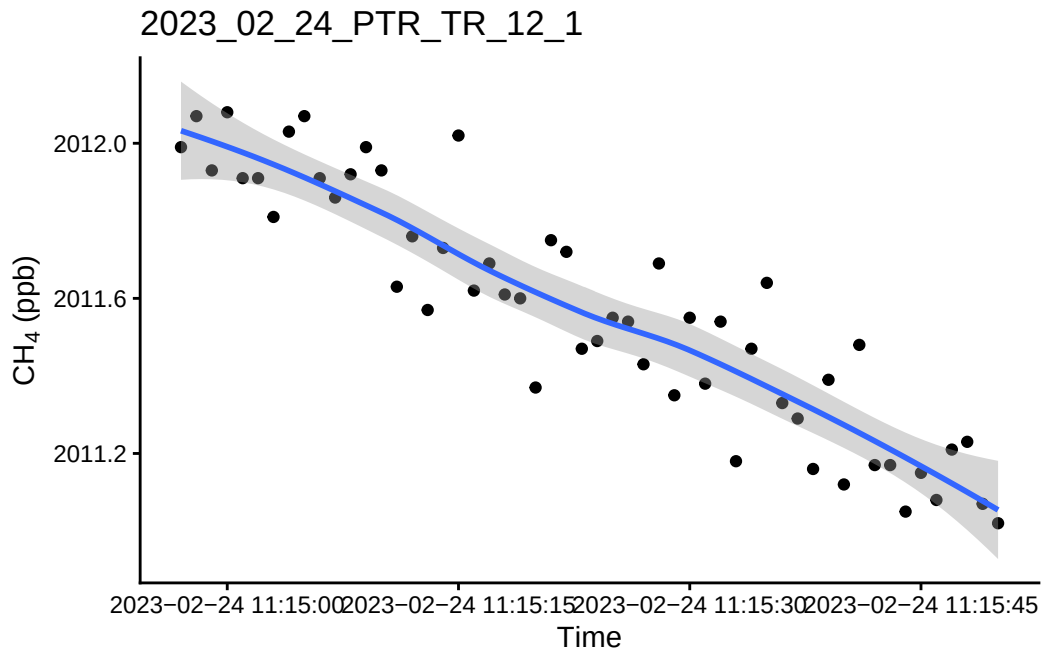


1.1 Look at the really high or really low ones to see if there is a timing issue or something

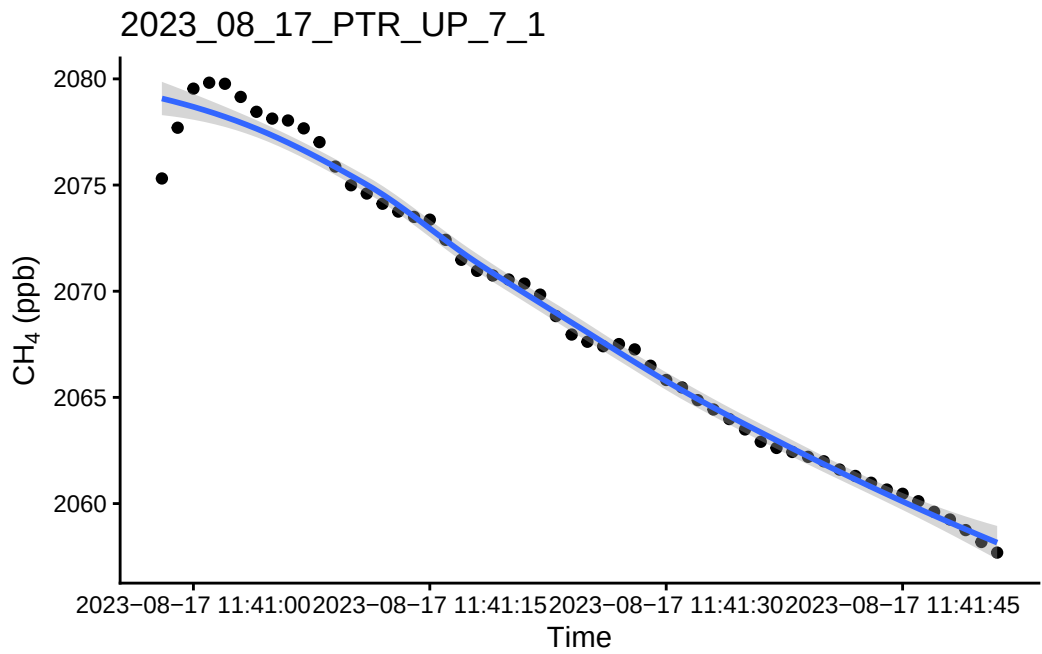
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



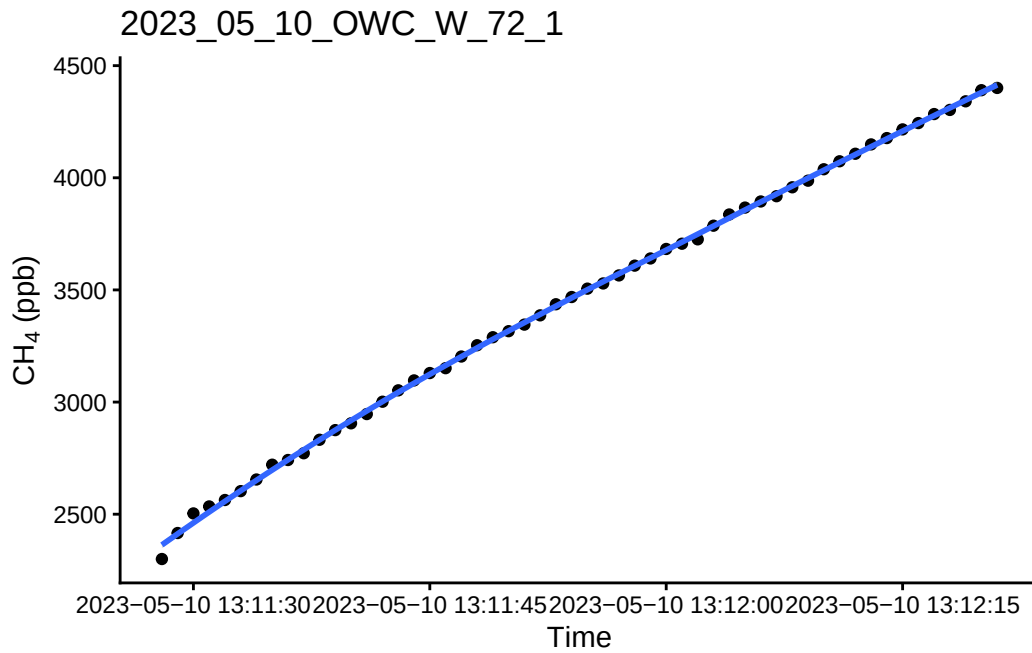
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



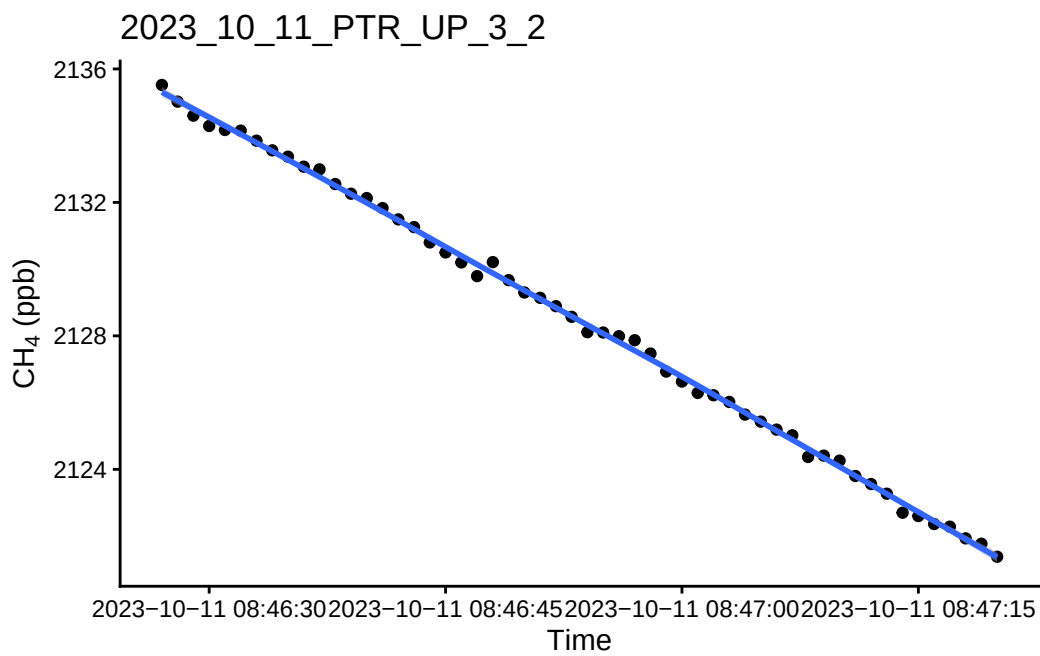
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

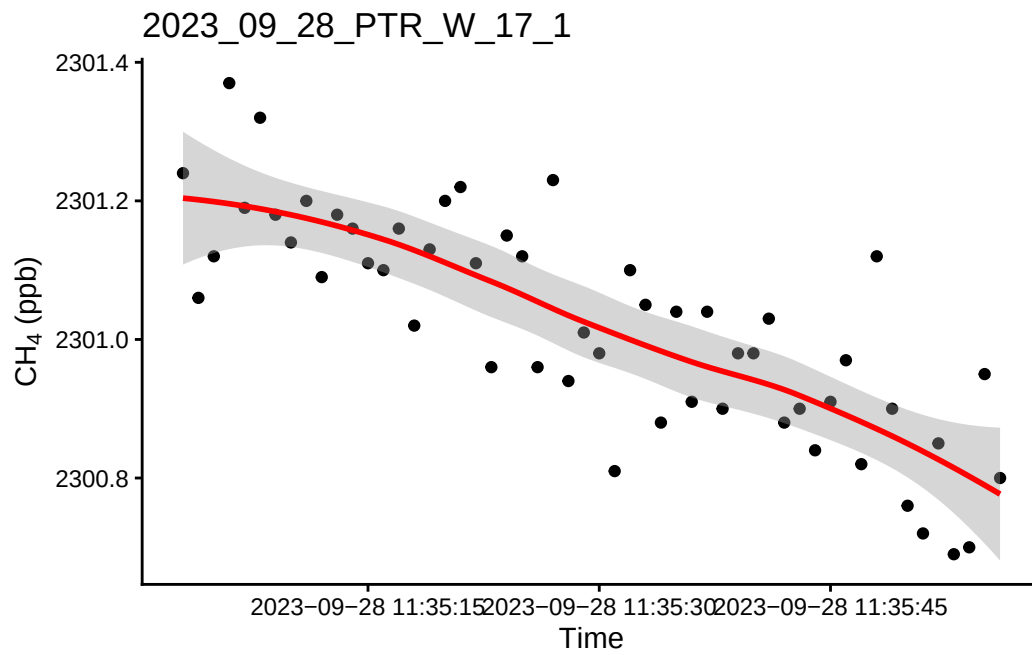


``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

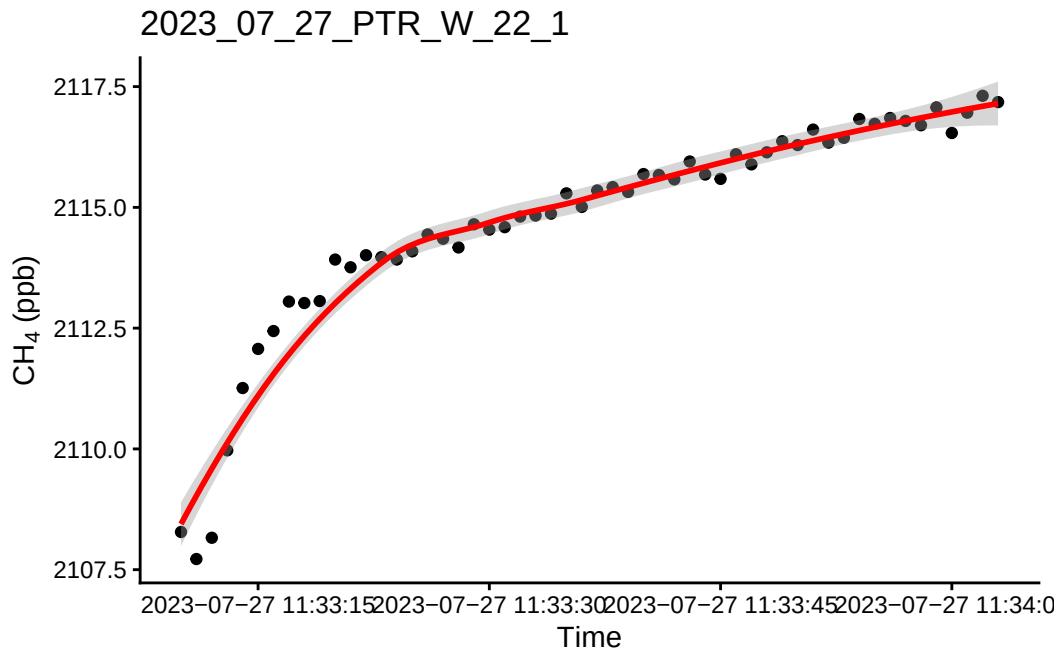


1.2 Check all fluxes

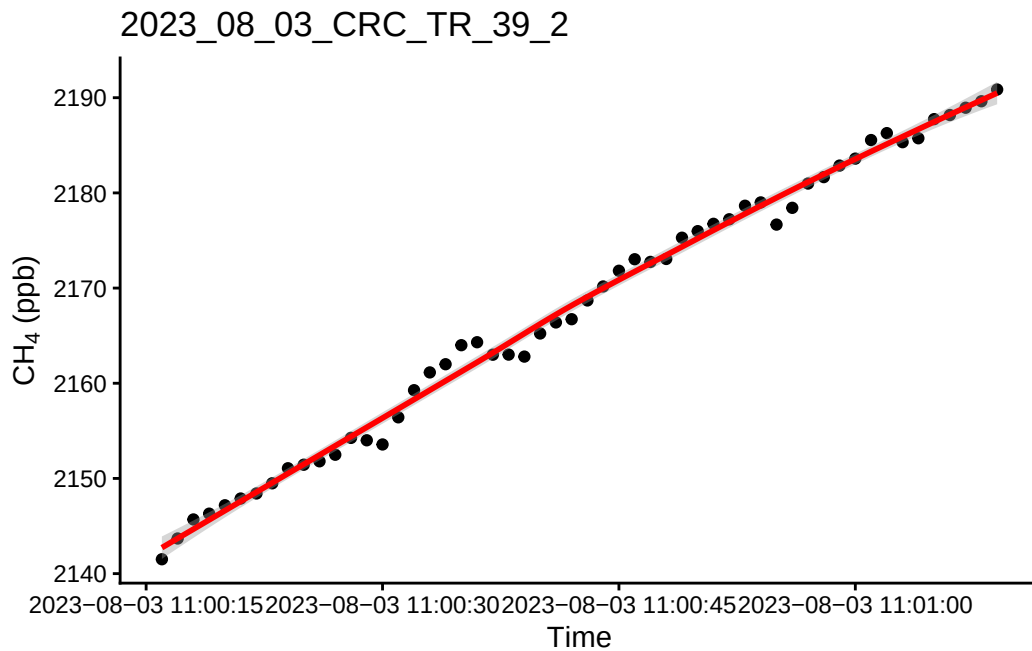
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



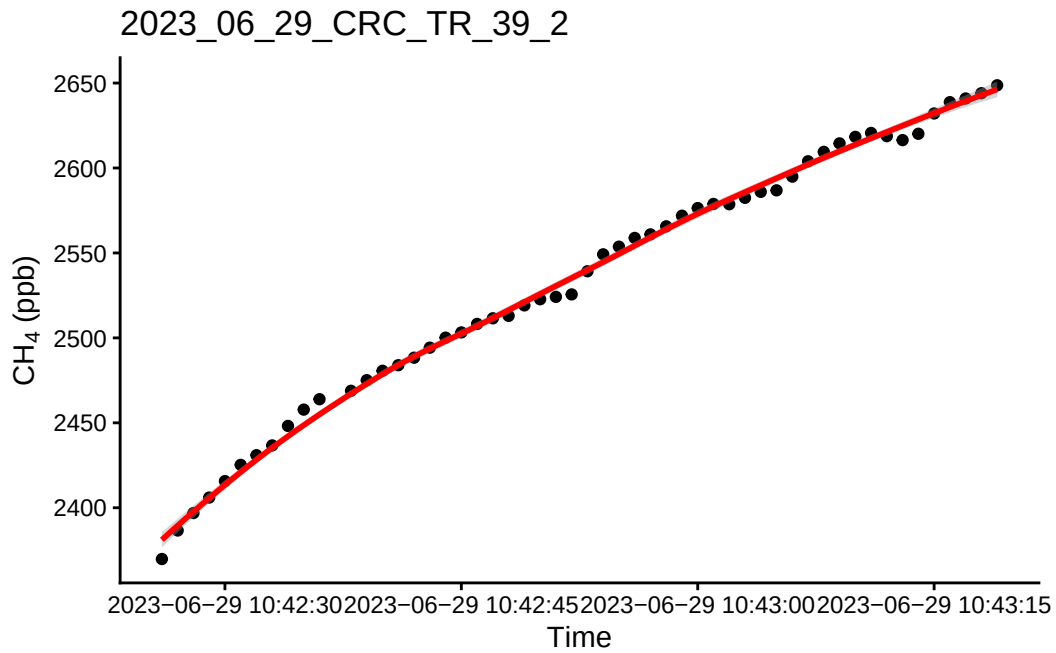
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



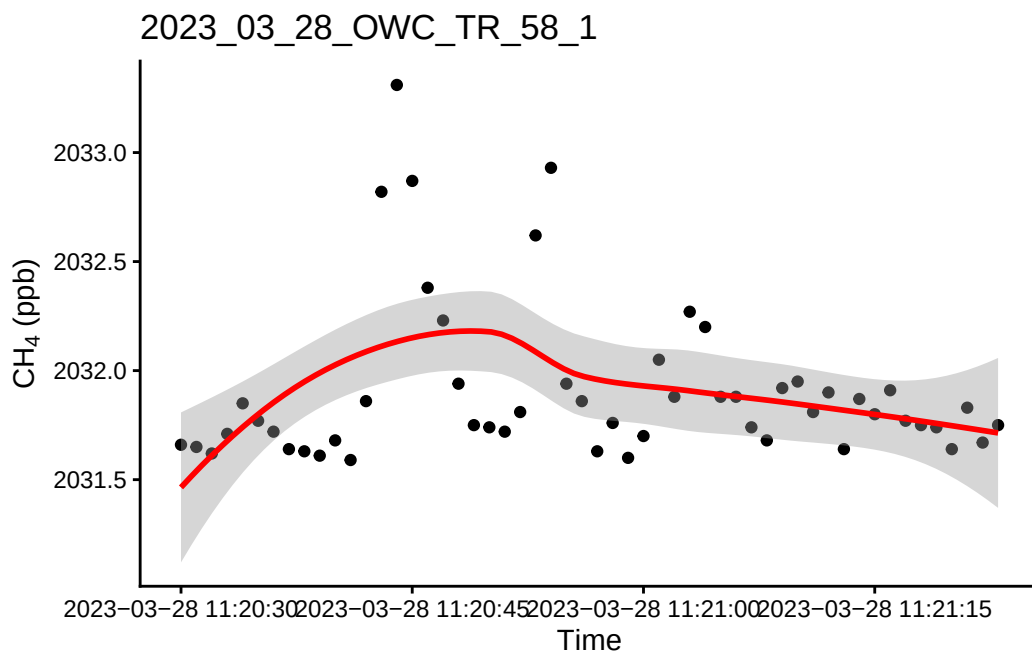
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



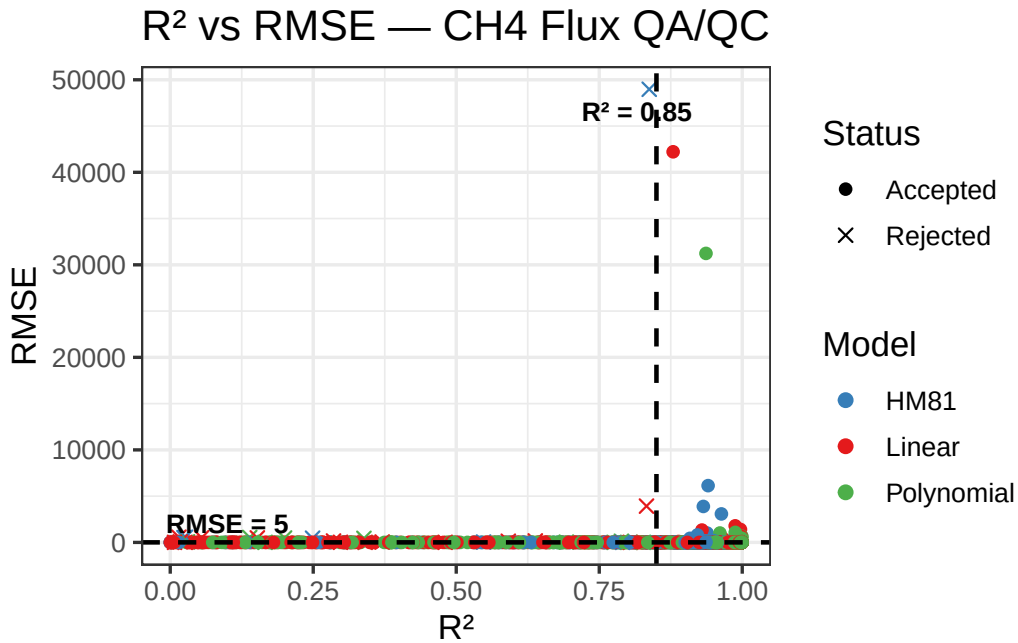
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



1.3 QAQC - Flags

Table 1: QA/QC CH4 Flux Summary

Metric	Value
Total observations	867
Rejections	
Rejected — poor model fit (n)	9
Rejected — model (%)	1
Rejected - [CH4] (n)	15
Rejected — [CH4] (%)	1
Model performance	
Passed — Linear model (n)	852
Passed — HM81 model (n)	451
Passed — Polynomial model (n)	858



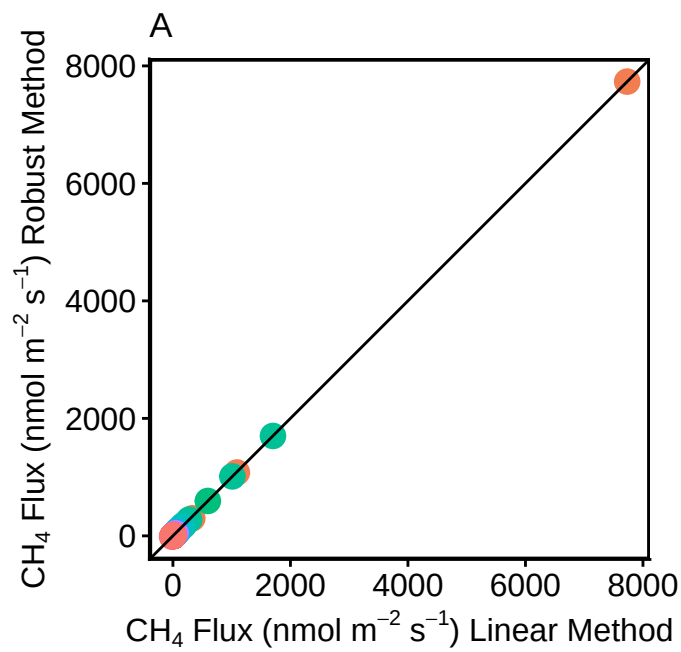
1.4 Remove any fluxes that do not fit criteria

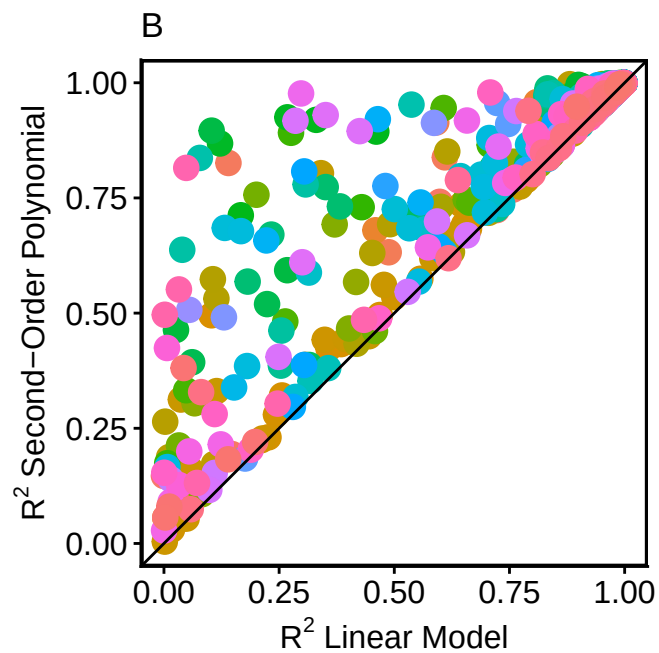
	Plot_ID	TIMESTAMP	HM81_AIC	HM81_flux.estimate
1	2023_01_05_OWC_TR_58_1	2023-01-05 11:06:46	222.8080	-0.60439179
2	2023_01_05_OWC_TR_58_2	2023-01-05 11:08:12	186.7132	-0.41206877

3	2023_01_05_OWC_TR_60_1	2023-01-05	11:21:11	NA	NA
4	2023_01_05_OWC_TR_60_2	2023-01-05	11:22:34	156.1166	-0.19504349
5	2023_01_05_OWC_TR_61_1	2023-01-05	11:27:49	172.9428	-0.09611557
6	2023_01_05_OWC_TR_61_2	2023-01-05	11:29:14	NA	NA
	HM81_p.value	HM81_r.squared	HM81_RMSE	lin_AIC	lin_flux.estimate
1	6.269688e-31	0.9397137	2.158799	152.9344	-0.5924959218
2	3.727399e-31	0.9410043	1.504716	162.4619	-0.4129079742
3	NA	NA	NA	128.6229	0.0006416185
4	1.128174e-13	0.6863264	1.108091	143.2382	-0.1169192217
5	1.590632e-04	0.2592945	1.311144	168.4159	-0.0588229511
6	NA	NA	NA	127.7154	-0.0913932579
	lin_flux.std.error	lin_int.estimate	lin_int.std.error	lin_p.value	
1	0.010519076	16450.12	0.3454520	1.665288e-45	
2	0.011570574	16431.66	0.3799837	3.243351e-36	
3	0.011845482	16448.34	0.3480839	9.570600e-01	
4	0.009547017	16454.71	0.3135290	2.225538e-16	
5	0.012280419	16468.10	0.4032954	1.642014e-05	
6	0.008174343	16475.01	0.2684497	5.791785e-15	
	lin_r.squared	lin_RMSE	poly_AIC	poly_r.squared	poly_RMSE
1	9.850960e-01	1.0733840	136.1049	0.9901739	0.8902985
2	9.636775e-01	1.1806806	147.6966	0.9750435	0.9997182
3	6.985037e-05	0.9977665	125.6517	0.1465829	0.9445379
4	7.575526e-01	0.9741935	145.0097	0.7681212	0.9732146
5	3.234091e-01	1.2531143	168.5142	0.3741989	1.2310834
6	7.225492e-01	0.8341235	131.5064	0.7237064	0.8502848
	rob_converged	rob_flux.estimate	rob_flux.std.error	rob_RMSE	
1	TRUE	-0.590749481	0.010016803	0.8677538	
2	TRUE	-0.419000426	0.011353684	0.9777928	
3	TRUE	0.002809075	0.012501125	0.9467892	
4	TRUE	-0.114050213	0.008415038	0.7184101	
5	TRUE	-0.057399172	0.013026729	1.3274822	
6	TRUE	-0.092496022	0.008228596	0.8547605	
	TIMESTAMP_min	TIMESTAMP_max	soilp_c	soilp_m	soilp_t
1	2023-01-05 11:06:22	2023-01-05 11:07:11	0.00500000	0.2020000	6.50
2	2023-01-05 11:07:48	2023-01-05 11:08:37	0.00500000	0.2020000	6.50
3	2023-01-05 11:20:50	2023-01-05 11:21:33	0.00422449	0.2355510	6.20
4	2023-01-05 11:22:10	2023-01-05 11:22:59	0.00500000	0.2340000	6.20
5	2023-01-05 11:27:25	2023-01-05 11:28:14	0.00400000	0.2463455	6.20
6	2023-01-05 11:28:50	2023-01-05 11:29:39	0.00400000	0.2470000	6.12
	CH4_initial	CH4_lin_pass	CH4_HM81_pass	CH4_poly_pass	CH4_init_conc
1	2018.27	TRUE	TRUE	TRUE	FALSE
2	2017.63	TRUE	TRUE	TRUE	FALSE
3	2021.23	TRUE	FALSE	TRUE	FALSE

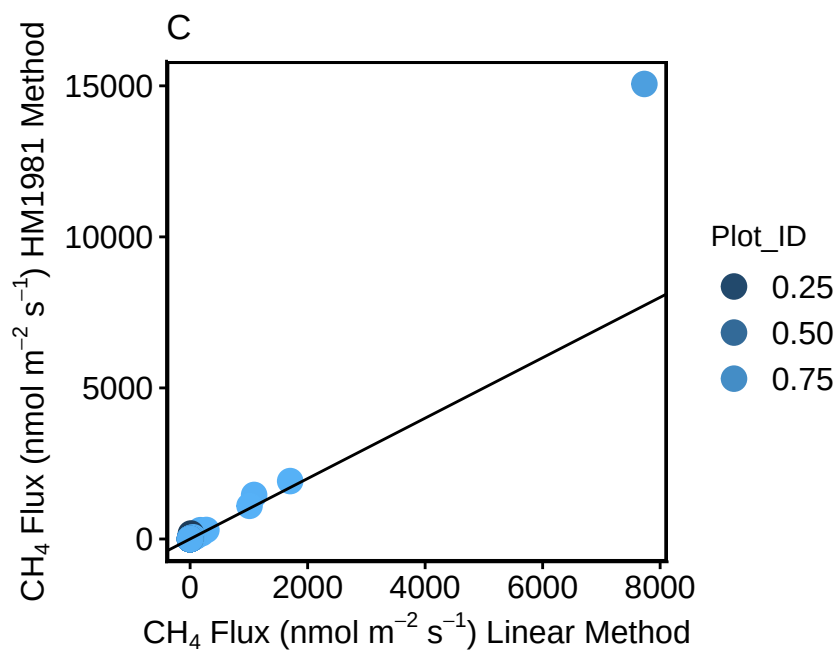
4	2020.74	TRUE	TRUE	TRUE	FALSE
5	2024.15	TRUE	TRUE	TRUE	FALSE
6	2023.64	TRUE	FALSE	TRUE	FALSE
	CH4_rejected_model	CH4_rejected_manual	CH4_flux_rejected		
1	FALSE	FALSE	FALSE		
2	FALSE	FALSE	FALSE		
3	FALSE	FALSE	FALSE		
4	FALSE	FALSE	FALSE		
5	FALSE	FALSE	FALSE		
6	FALSE	FALSE	FALSE		

1.5 Do some visualization of QAQC plots to assess fluxes





Warning: Removed 410 rows containing missing values or values outside the scale range (``geom_point()``).



1.6 Read out final dataframe and export for use elsewhere

```
[1] "TR" "UP" "W"
```

```
[1] 58 60 61 63 64 50 51 52 55 56 65 66 68 71 72 11 12 14 15 9 2 3 4 6 7  
[26] 17 18 20 22 23 NA 62 67 36 37 25 26 27 29 30 43 46 8 33 34 39 41 53 42 45  
[51] 48 70 35 38 40 47 59 32 69 28 1 21 44
```

```
[1] 1 2
```